

## Section 1: Technical Details

### 1.1 Preprocessing

The processing consists on several steps

- **Target Variable Transformation:** We transformed the target column by taking its natural logarithm (log transformation) in the training dataset.
- **Dropping Columns:** Several columns, such as 'PID', 'Street', 'Utilities', 'Condition\_2', 'Roof\_Mat', 'Heating', 'Pool\_QC', 'Misc\_Feature', 'Low\_Qual\_Fin\_SF', 'Pool\_Area', 'Longitude' and 'Latitude', are dropped from the dataset as they are not considered relevant for the model.
- **Numeric Data Clipping:** Numeric columns are clipped at a specified 95<sup>th</sup> quantile. These quantiles are computed using only training data.
- **Categorical Data Encoding:** The categorical columns that were not removed are one-hot encoded. The categories are defined using only training data.
- **Missing Value Handling:** Any missing values are filled with zeros in the dataset.

The same transformation is applied for the testing data. In this case we do not have access to the target, so “Target Variable Transformation” is ignored. Let’s keep in mind that in the case of “Numeric Data Clipping” and “Categorical Data Encoding” we use parameters already fitted on training data. The other processing steps do not require prior fit.

### 1.2 Modeling

Two machine learning models are initialized:

**Linear model:** It's a Ridge Regression trained using alpha min from a cross validation. Specifically, we used RidgeCV from sklearn, with alpha taking the following values: 0.1, 1.0 and 10.0 (default in sklearn).

**Tree model:** It's an Boosted tree regressor. Specifically, we use XGBRegressor from xgboost library with the following hyperparameters:

- Number of estimators = 500
- Maximum depth of the trees = 6
- Eta = 0.05
- Subsample = 0.5
- Objective metric = Squared error
- Random state (seed) = 0

**Model Prediction:** Both the linear model and tree model are used to make predictions on the preprocessed test data. The predictions are exponentiated to revert the log transformation done earlier on the target variable.

## Section 2: Performance metrics

### Windows 11 Pro

- Processor: 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz 1.38 GHz
- RAM: 16GB

|    | Preprocessing<br>training time (s) | Performance<br>(Linear) | Time<br>training (s) | Performance<br>(Tree) | Time<br>training (s) | Total<br>time (s) |
|----|------------------------------------|-------------------------|----------------------|-----------------------|----------------------|-------------------|
| 1  | 0.09                               | 0.1206                  | 0.14                 | 0.1154                | 11.00                | 11.23             |
| 2  | 0.08                               | 0.1166                  | 0.14                 | 0.1219                | 11.30                | 11.52             |
| 3  | 0.10                               | 0.1139                  | 0.14                 | 0.1125                | 10.84                | 11.08             |
| 4  | 0.10                               | 0.1148                  | 0.16                 | 0.1182                | 10.32                | 10.59             |
| 5  | 0.08                               | 0.1088                  | 0.14                 | 0.1104                | 10.51                | 10.74             |
| 6  | 0.11                               | 0.1314                  | 0.16                 | 0.1279                | 10.88                | 11.15             |
| 7  | 0.08                               | 0.1307                  | 0.15                 | 0.1338                | 10.63                | 10.86             |
| 8  | 0.09                               | 0.1266                  | 0.15                 | 0.1260                | 9.97                 | 10.20             |
| 9  | 0.09                               | 0.1293                  | 0.15                 | 0.1335                | 9.86                 | 10.10             |
| 10 | 0.10                               | 0.1220                  | 0.16                 | 0.1202                | 11.01                | 11.27             |